2.7

1. According to the VSEPR model, the progressive decrease in the bond angles in the series of molecules CH_4, NH_3, and H_2O is best accounted for by the

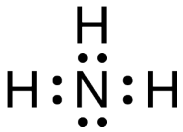
(A) increasing strength of the bonds

(B) increasing electronegativity of the central atom

(C) increasing number of unshared pairs of electrons ✓

(D) decreasing repulsion between hydrogen atoms

2.



Based on the Lewis diagram for NH_3, shown above, the H-N-H bond angle is closest to which of the following?

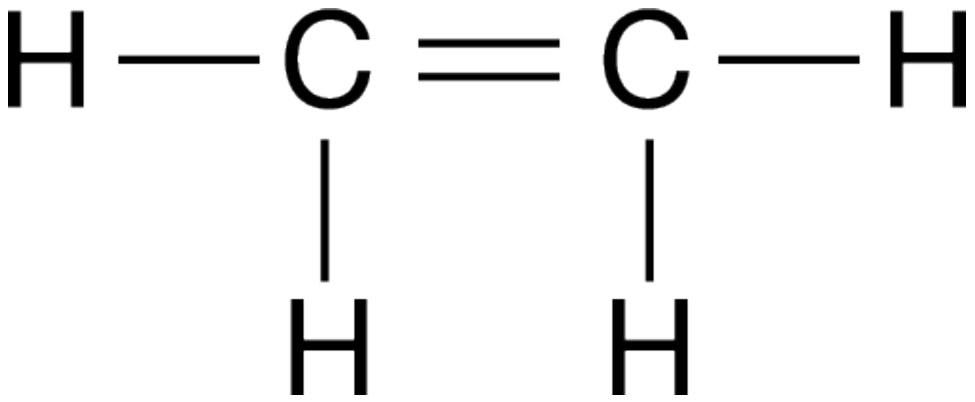
(A) 60°

(B) 90°

(C) 109.5° ✓

(D) 120°

3.



A Lewis diagram for the molecule C_2H_4 is shown above. In the actual C_2H_4 molecule, the H-C-H bond angles are closest to

(A) 90°

(B) 109.5°

(C) 120° ✓

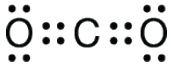
(D) 180°

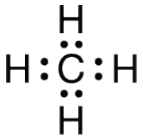
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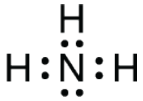
Answer C

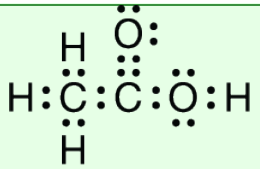
Correct. With three electron domains around each C atom, their geometry is trigonal planar (which minimizes repulsion among the three electron pairs). Therefore the H-C-H bond angle is approximately 120°.

4. Which of the following complete Lewis diagrams represents a molecule containing a bond angle that is closest to 120°?

(A) 

(B) 

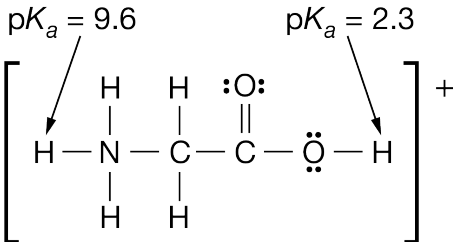
(C) 

(D) 

Answer D

Correct. The carbon atom attached to the oxygen atoms exhibits sp^2 hybridization, thus the bond angles around it are close to 120°. Another explanation is, the carbon atom attached to the two oxygen atoms is surrounded by three electron domains, which, according to VSEPR theory, will repel one another to produce a trigonal planar geometry around that atom.

2.7



The structural formula of the glycinium cation is shown above. Arrows indicate the $\mathrm{p}K_a$ values for the labile protons in the molecule.

5. What is the approximate $\mathrm{H} - \mathrm{O} - \mathrm{C}$ bond angle in the glycinium cation?

(A) 180°

(B) 120°

(C) 105° ✓

(D) 90°

Answer C

Correct. The oxygen atom in the $\mathrm{H} - \mathrm{O} - \mathrm{C}$ bond has two lone pairs of electrons, so the geometry of the electron domains around the oxygen atom is roughly tetrahedral. The bond angle is thus close to 109°.

6. Which of the following is true about the geometry of the glycinium cation?

(A) The leftmost C atom and all the atoms directly bonded to it lie in the same plane.

(B) Both C atoms and both O atoms lie in the same plane. ✓

(C) The $\mathrm{N} - \mathrm{C} - \mathrm{C}$ bond angle is 180°.

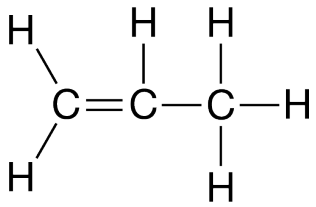
(D) The geometry around the N atom is planar.

Answer B

Correct. The carbon atom bonded to two oxygen atoms and another carbon atom does not have any nonbonding pairs of electrons, thus the central carbon atom has three electron domains. According to VSEPR theory, all four atoms are in the same plane (sp^2 hybridization).

2.7

7.



Which of the following correctly compares the strength of the two carbon-to-carbon bonds in the molecule represented in the Lewis diagram shown above?

(A) The carbon-to-carbon bond on the left is stronger because it is a double bond. ✓

(B) The carbon-to-carbon bonds are the same strength because the C-C-C bond angle is 180°.

(C) The carbon-to-carbon bonds are the same strength because they are both bonds between carbon atoms.

(D) The carbon-to-carbon bond on the right is stronger because there are more hydrogen atoms attached to the rightmost carbon atom than to the leftmost carbon atom.

Answer A

Correct. Double bonds between two atoms are always stronger than single bonds between those same two atoms because in a double bond there are more electrons between the two positive nuclei of the atoms, which results in a stronger bond.

Reaction 1: $CaC_2(s) + 2\ H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$

Reaction 2: $NaOCl(aq) + 2\ HCl(aq) \rightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$

Reaction 3: $C_2H_2(g) + Cl_2(g) \rightarrow C_2H_2Cl_2(g)$

8. When Reaction 3 occurs, does the hybridization of the carbon atoms change?

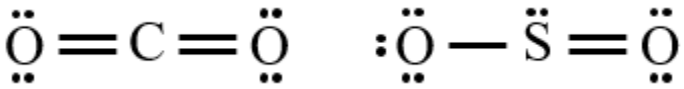
(A) Yes; it changes from *sp* to sp^2. ✓

(B) Yes; it changes from *sp* to sp^3.

(C) Yes; it changes from sp^2 to sp^3.

(D) No; it does not change

2.7

9. 

Lewis electron-dot diagrams for CO_2 and SO_2 are given above. The molecular geometry and polarity of the two substances are

(A) the same because the molecular formulas are similar

(B) the same because C and S have similar electronegativity values

(C) different because the lone pair of electrons on the S atom make it the negative end of a dipole

(D) different because S has a greater number of electron domains (regions of electron density) surrounding it than C has ✓

Answer D

This option is correct. Both the C-O and S-O bonds are polar, due to the different electronegativities of the bonded atoms. However, the CO_2 molecule is linear and so the bond dipoles cancel and the molecule has no net dipole. In SO_2, the molecule is bent and so the bond dipoles do not cancel and a net dipole is present. The polarity of the two substances differs because of the differences in geometry, which is a direct consequence of the number of electron domains around the central atom (2 in CO_2 and 3 in SO_2), as stated in this option.

Refer to the following diatomic species

(A) Li_2

(B) B_2

(C) N_2

(D) O_2

(E) F_2

10. Contains 1 sigma (σ) and 2 pi (π) bonds

(A) B_2

(B) N_2 ✓

(C) O_2

(D) F_2

2.7

11. Has a bond order of 2

(A) B_2

(B) N_2

(C) O_2 ✓

(D) F_2

12. Has the largest bond-dissociation energy

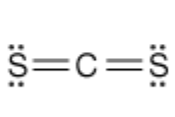
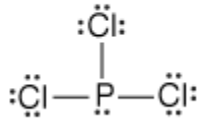
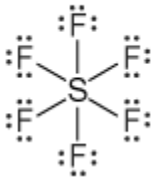
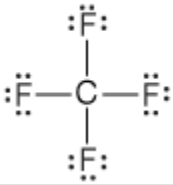
(A) B_2

(B) N_2 ✓

(C) O_2

(D) F_2

13.

CS_2	PCl_3	SF_6	CF_4
			

Based on the diagrams shown above, which of the following would have the greatest dipole moment, and why?

(A) CS_2, because the double bonds create an area of higher electron density around the C atom.

(B) PCl_3, because the Cl atoms draw electron density away from the P atom and the bond dipoles do not cancel one another. ✓

(C) SF_6, because it has the greatest number of polar bonds.

(D) CF_4, because the large electronegativity difference between C and F results in very polar bonds.

Answer B

Correct. PCl_3 is predicted by Valence Shell Electron Pair Repulsion (VSEPR) theory to have a trigonal pyramidal molecular shape. The central P atom has four electron domains around it, giving a tetrahedral electronic geometry with one of the positions occupied by a lone pair of electrons. The bond dipoles would not cancel, and PCl_3 would have a nonzero dipole moment.

2.7

Directions: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) SO_2

(C) N_2

(D) F_2

14. Is the molecule in which the intramolecular forces are strongest

(A) $BeCl_2$
(B) SO_2
(C) N_2 ✓
(D) F_2

15. Is a polar molecule

(A) $BeCl_2$
(B) SO_2 ✓
(C) N_2
(D) F_2

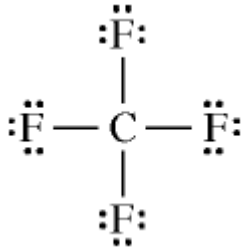
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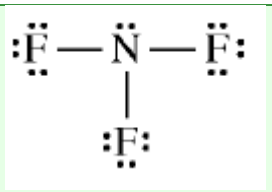
16. The geometry of the SO_3 molecule is best described as

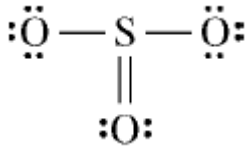
(A) trigonal planar ✓
(B) trigonal pyramidal
(C) bent
(D) tetrahedral

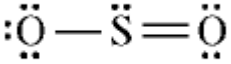
17. Which of the following Lewis electron-dot diagrams represents the molecule that contains the smallest bond angle?

2.7

(A) 

(B) 

(C) 

(D) 

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The question refers to the following species.

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4

18. Has a trigonal-pyramidal molecular geometry

(A) H_2O
(B) NH_3 ✓
(C) BH_3
(D) CH_4

19. Is predicted to have the largest bond angle

2.7

(A) H_2O
(B) NH_3
(C) BH_3 ✓
(D) CH_4

---

20. Which of the following has molecules with a pyramidal shape?

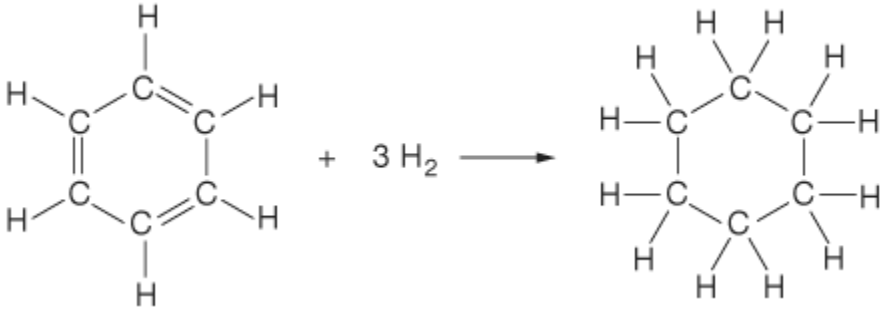
(A) $NH_3(g)$ ✓
(B) $BH_3(g)$
(C) $H_2S(g)$
(D) HBr(g)

21. H - C ≡ C - H

What is the hybridization of the carbon atoms in a molecule of ethyne, represented above?

(A) *sp* ✓
(B) sp^2
(C) sp^3
(D) dsp^2

22.



In the reaction represented above, which of the following sets correctly identifies the hybridization of the C atoms before and after the reaction occurs?

Set	Before	After
A	sp^2	sp^2
B	sp^2	sp^3
C	sp^3	sp^2
D	sp^3	sp^3

2.7

(A) Set A

(B) Set B ✓

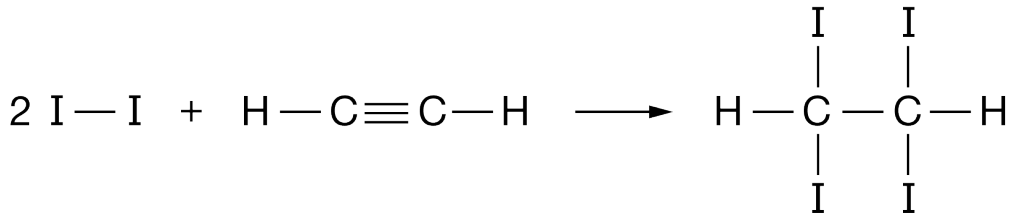
(C) Set C

(D) Set D

Answer B

Correct. In the C_6H_6 molecule, each carbon atom is surrounded by three electron domains in a trigonal planar geometry, and the hybridization of the orbitals of each carbon atom is sp^2. In the C_6H_{12} molecule, each carbon atom is surrounded by four electron domains in a tetrahedral geometry, and the hybridization of the orbitals of each carbon atom is sp^3.

23. The reaction between I_2 and C_2H_4 is represented by the following equation.



How does the hybridization of the carbon atoms change from reactants to products?

(A) sp to sp^2

(B) sp to sp^3 ✓

(C) sp^2 to sp

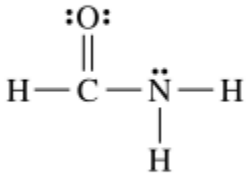
(D) sp^3 to sp

Answer B

Correct. Each C atom in C_2H_4 has two electron domains, so they have sp hybridization. Each C atom in $C_2H_2I_4$ has four electron domains, so they have sp^3 hybridization. The hybridization of the carbon atoms therefore changes from sp to sp^3.

2.7

24.



The complete Lewis electron-dot diagram for $HCONH_2$ is shown above. Identify the hybridization of the valence orbitals for the carbon atom and the approximate hydrogen-carbon-nitrogen bond angle according to VSEPR theory.

(A) sp hybridized and 180°

(B) sp^2 hybridized and 120° ✓

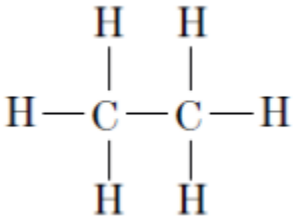
(C) sp^2 hybridized and 180°

(D) sp^3 hybridized and 120°

Answer B

Correct. The carbon atom has three electron domains: the $C-H$ bond, the $C-N$ bond, and the $C=O$ bond. According to VSEPR theory, three domains around an atom achieve maximum separation and greatest stability if the domains are arranged in a trigonal planar fashion, with 120° bond angles. When there are three electron domains, the valence orbitals are sp^2 hybridized.

25.

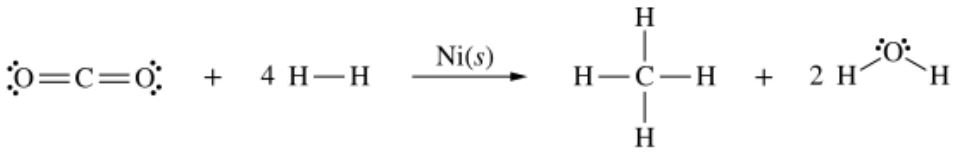


The hybridization of the carbon atoms in the molecule represented above can be described as

2.7

(A) *sp*
(B) sp^2
(C) sp^3 ✓
(D) dsp^2

In the gas-phase reaction represented by the following equation, methane (CH_4) and water (H_2O) are produced from carbon dioxide (CO_2) and hydrogen (H_2) at high temperature and pressure.



26. Which of the following identifies the hybridization of the valence orbitals of the central atoms of the methane and water molecules?

	CH_4	H_2O
A	sp^2	sp
B	sp^2	sp^2
C	sp^3	sp^2
D	sp^3	sp^3

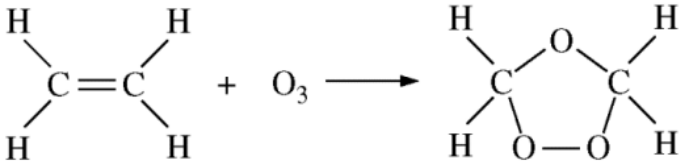
(A) A
(B) B
(C) C
(D) D ✓

Answer D

Correct. The hybridization of the valence orbitals of the central atoms of CH_4 and H_2O are both sp^3 because both have four electron domains around the central atom.

2.7

27.



In the reaction represented above, what is the hybridization of the C atoms before and after the reaction occurs?

(A)

Before	After
sp	sp^2

(B)

Before	After
sp	sp^3

(C)

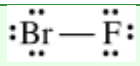
Before	After
sp^2	sp

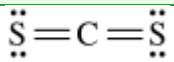
(D) ✓

Before	After
sp^2	sp^3

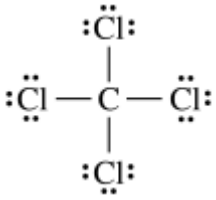
28. Which of the following molecules has the largest dipole moment?

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(A)  ✓

(B) 

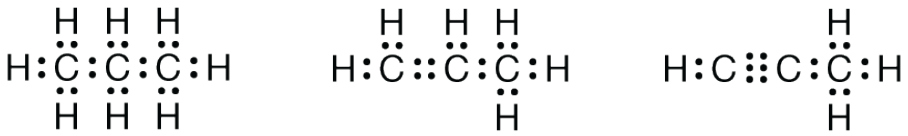
(C) 

(D) 

Answer A

Correct. The molecule with the largest dipole moment has bond dipoles that do not cancel as a result of the molecular shape. The BrF molecule has a bond dipole and the molecule is not symmetrical, so the bond dipoles do not cancel.

29.



Molecule 1 Molecule 2 Molecule 3

Lewis diagrams of molecules of three different hydrocarbons are shown above. Which of the following claims about the molecules is best supported by the diagrams?

(A) All the atoms in molecule 1 lie in one plane.

(B) All the molecules have the same empirical formula.

(C) The C-C-C bond angle in molecule 2 is close to 180°.

(D) The strongest carbon-to-carbon bond occurs in molecule 3. ✓

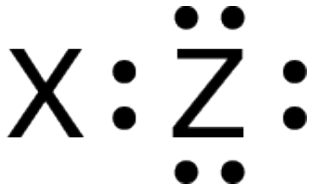
Answer D

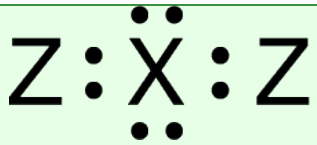
Correct. Molecule 3 contains a triple bond between two carbon atoms. Between two of the same kind of

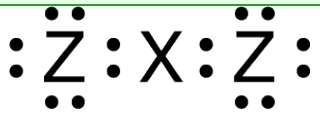
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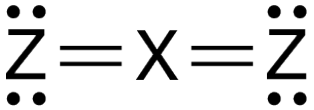
atom, triple bonds are stronger than double bonds, which in turn are stronger than single bonds.

30. In the following diagrams, elements are represented by X and Z, which form molecular compounds with one another. Which diagram represents a molecule that has a bent molecular geometry?

(A) 

(B)  ✓

(C) 

(D) 

Answer B

Correct. There are four electron domains around the central atom X, and their geometry is tetrahedral, which minimizes the repulsion between the four electron pairs. The Z-X-Z bond angle is approximately 109.5°, and the molecular geometry is bent.

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

31. The molecule with the largest dipole moment

2.7

(A) CO_2

(B) H_2O ✓

(C) C_2H_4

(D) PH_3

32. The molecule that has trigonal pyramidal geometry

(A) CO_2

(B) H_2O

(C) C_2H_4

(D) PH_3 ✓

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33. Which of the following molecules contains exactly three sigma (σ) bonds and two pi (π) bonds?

(A) C_2H_2 ✓

(B) CO_2

(C) HCN

(D) SO_3

34. Of the following molecules, which has the largest dipole moment?

(A) CO

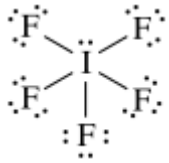
(B) CO_2

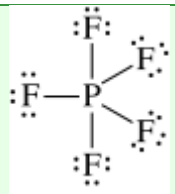
(C) HF ✓

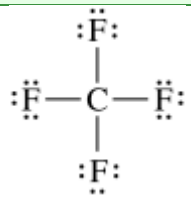
(D) F_2

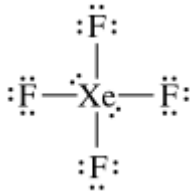
35. Which of the following represents a molecule with a trigonal bipyramidal geometry?

2.7

(A) 

(B)  ✓

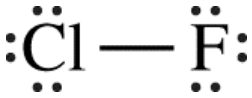
(C) 

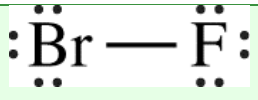
(D) 

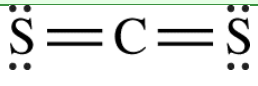
Answer B

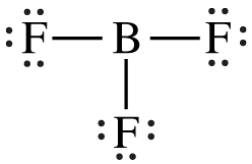
Correct. The Lewis diagram for a molecule having trigonal bipyramidal geometry must have five bonding pairs and no nonbonding pairs of electrons around the central atom. This response has the required arrangement of bonding and nonbonding pairs around the central P atom.

36. Which of the following Lewis electron-dot diagrams represents the molecule that is the most polar?

(A) 

(B)  ✓

(C) 

(D) 

2.7

37. Which of the following molecules is nonpolar but has polar covalent bonds?

(A) H_2O_2

(B) H_2O

(C) CCl_4 ✓

(D) CH_2Cl_2

38. Which of the following arranges the molecules N_2, O_2, and F_2 in order of their bond enthalpies, from least to greatest?

(A) $F_2 < O_2 < N_2$ ✓

(B) $O_2 < N_2 < F_2$

(C) $N_2 < O_2 < F_2$

(D) $N_2 < F_2 < O_2$

39. Pi (π) bonding occurs in each of the following species EXCEPT

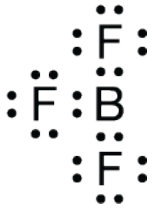
(A) CO_2

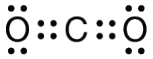
(B) CN^-

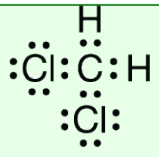
(C) C_6H_6

(D) CH_4 ✓

40. Which of the following Lewis diagrams represents a molecule that is polar?

(A) 

(B) 

(C)  ✓

(D) 

Answer C

Correct. In the CH_2Cl_2 molecule, the H and Cl atoms are in a tetrahedral arrangement around the

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central C atom. The two electronegative chlorine atoms create a partial negative charge on their side of the carbon atom, leaving a partial positive charge on the opposite (hydrogen) side of the carbon atom. This separation of charge constitutes a dipole, and the molecule is polar.

An equimolar mixture of SO_2 and O_2 is placed in a rigid container at $298\ K$. The initial pressure in the container is $0.20\ atm$. A catalyst is added to the mixture, and the reaction represented below occurs.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \qquad \Delta H^\circ = -200\ kJ/mol_{rxn}$$

41. Which of the following gives the shape of the SO_2 molecule and best helps explain why it has that shape?

(A) It is linear, because there are two electron domains around the sulfur atom.

(B) It is linear, because there are three electron domains around the sulfur atom.

(C) It is bent, because there are two electron domains around the sulfur atom.

(D) It is bent, because there are three electron domains around the sulfur atom. ✓

Answer D

Correct. A Lewis electron-dot diagram for SO_2 can be drawn with two equivalent resonance structures. S is the central atom, and there is one single bond and one double bond, with resonance between the two bonds. The S atom has one nonbonding pair of electrons. Since there are two bonds and one nonbonding pair of electrons on the S atom, the molecule is bent.

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42. The BF_3 molecule is nonpolar, whereas the NF_3 molecule is polar. Which of the following statements accounts for the difference in polarity of the two molecules?

(A) In NF_3, each F is joined to N with multiple bonds, whereas in BF_3 , each F is joined to B with single bonds.

(B) N — F bonds are polar, whereas B — F bonds are nonpolar.

(C) NF_3 is an ionic compound, whereas BF_3 is a molecular compound.

(D) Unlike BF_3, NF_3 has a nonplanar geometry due to an unshared pair of electrons on the N atom. ✓

43. Which of the molecules represented below contains carbon with sp^2 hybridization?

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(A) CH_2Cl_2

(B) C_2H_6

(C) $C_2H_2Cl_2$ ✓

(D) $C_2H_4Cl_2$

44. Which of the following has a zero dipole moment?

(A) HCN

(B) SO_2

(C) NO_2

(D) PF_5 ✓

45. Which of the following species is NOT planar?

(A) CO_3^{2-}

(B) NO_3^-

(C) ClF_3

(D) PCl_3 ✓